Dhairya Shah

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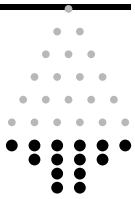

EDUCATION

Ahmedabad University — Class of 2028
B.Tech, Computer Science and Engineering (3rd year) — CGPA: 3.25/4.0

PROJECTS

GestureOS | Python, MediaPipe, OpenCV, CustomTkinter — github.com/Dhairya233/GestureOS

Control a Windows PC with your bare hands, through a webcam.

- **Built a Windows desktop app** that turns webcam hand tracking (MediaPipe Hand Landmarker, 21 landmarks per hand) into mouse, scroll, media, drawing and app-launch controls.
- **Designed a safety model:** the app starts disarmed, arms by a two-palm hold, disarms from any window with a global hotkey, and sends every OS event through one gateway that releases held keys on disarm.
- **Wrote 630 automated tests** that drive the real engine with generated hands and a fake input backend, so the project is tested without a camera, with CI on GitHub Actions.

Life-RPG Protocol | Next.js, TypeScript, Supabase/PostgreSQL — github.com/Dhairya233/life-rpg-protocol

A gamified productivity app where I found, then fixed, a privilege-escalation hole in my own database.

- **Built a gamified productivity web app** (Next.js 14, TypeScript, Supabase/PostgreSQL) with a focus-session monitor and reputation-weighted peer review of quest proof photos.
- **Audited my own database and found users could edit their own Aura, XP and role**; fixed it with column-level grants, revoked function rights and identity taken from auth.uid().
- **Wrote the attacks as tests:** 44 attack checks failed on the old schema and about 260 database checks now pass in CI, including a two-session race test against double payouts.

Browser Arcade | JavaScript, WebRTC, PeerJS — github.com/Dhairya233/browser-arcade

Two multiplayer board games where I found, then fixed, a hidden-information leak in my own game.

- **Built two multiplayer browser board games, Shadow Hunt and Game of Trades,** in plain JavaScript modules with WebRTC (PeerJS) and no build step.
- **Found a hidden-information leak in my own game** (26 of 26 snapshots exposed the hidden player's position) and fixed it with a host-authoritative design: per-seat views and validation of every action before it is applied.
- **Wrote a test that reads every message sent to every non-hidden seat across a whole online game**, plus a fuzz test with hostile payloads; 135 tests run on every push with GitHub Actions.

Fabric Defect Anomaly Detection | Python, scikit-learn — github.com/Dhairya233/Textile-Defect-ML-Model

A leak-aware study of how much your data split changes a fabric-defect detector's results.

- **Built a solo, leak-aware study of classical fabric-defect detection on the public AITEX dataset** (245 strips, 7 fabrics): 90 texture features per patch feeding a random forest, an SVM and four anomaly detectors, with a static results explorer.
- **Measured the cost of a careless split:** a random patch split overstated the random forest's patch-level average precision by 0.44 (95% interval 0.27 to 0.53) against splitting by whole strip, and by 0.64 when a whole fabric was held out; with a fabric held out every method scored close to chance.
- **Resampled whole strips (1,000 bootstrap resamples)** for intervals and wrote 75 passing tests, including checks that no strip sits on both sides of a split and that the numbers quoted in the README match the result tables.

RESEARCH EXPERIENCE

Institute of Manufacturing and Economy, Ahmedabad University — Snapshot published August 2026
Research Intern — ahduni.edu.in (Institute Insights: textiles sector snapshot)

- **Conducted the primary research** for the Institute's published sector snapshot "Textiles: Growing in Scale, Losing Relative Ground" (NIC Divisions 13 and 14, August 2026).
- Worked from **the Annual Survey of Industries, the Index of Industrial Production, DGCI&S trade statistics and CMIE Prowess** firm-level data, covering 2014-15 to 2024-25.
- **Findings include 61 percent growth in sector output from 2014-15 to 2023-24** alongside a lag behind overall manufacturing on the production index, and imports growing faster than exports.

LEADERSHIP & DESIGN

SAEINDIA Ahmedabad University Student Section — 2024 to 2025
Co-ordinating Committee Member and graphic designer

- Led one of the section's events as event head, responsible for running it from planning through to the end of the day.
- Designed graphics for the section's events and posts. Many of my designs were selected for the SAE Ahmedabad University Instagram page.

SKILLS

Languages Python JavaScript TypeScript SQL
Tools Next.js Supabase/PostgreSQL WebRTC MediaPipe OpenCV scikit-learn pytest GitHub Actions Git
Learning Data structures and algorithms C++ Data engineering
Spoken languages English Hindi Gujarati **Hobbies** Music

CERTIFICATES AND RECOGNITION

AWS Academy Cloud Foundations, AWS Academy
AWS Academy Graduate - Data Engineering, AWS Academy
Documents at dhairya233.github.io

Letter of Appreciation, SAEINDIA Ahmedabad University Student Section March 2025
Letter of Appreciation, Foundation Challenge, Ahmedabad University April 2025